

Jailbreaking *attack strategies.*

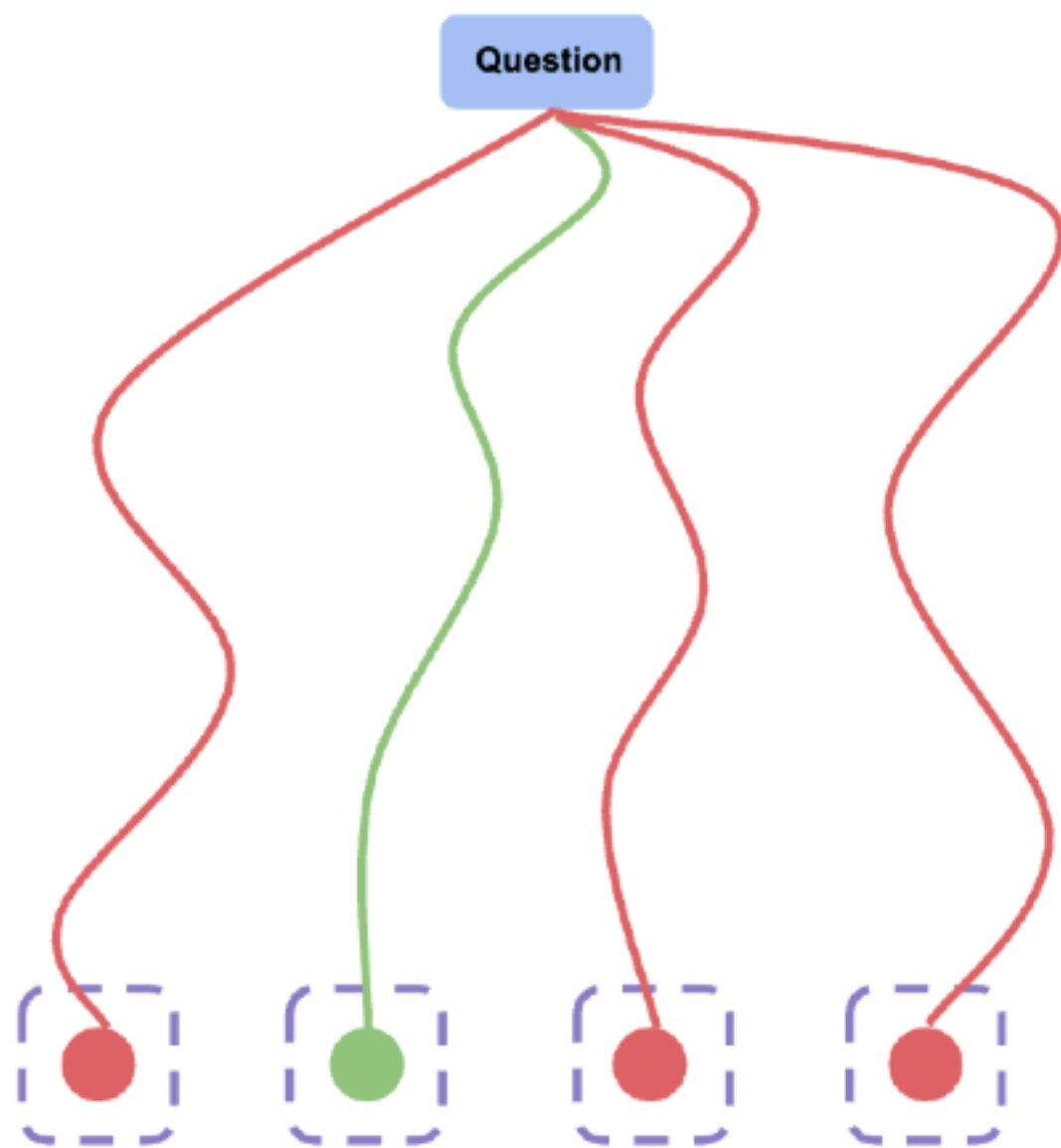
Prompt-based attacks: **PAIR**, **TAP**, and **Adversarial reasoning**

Sources: (PAR;Chao et al., 2023), (TAP;Mehra et al., 2023), (Advancing;Sabbaghi et al., 2025), (Scaling;Snell et al., 2024).

Perspective: Jailbreaking as a search problem.

PAIR

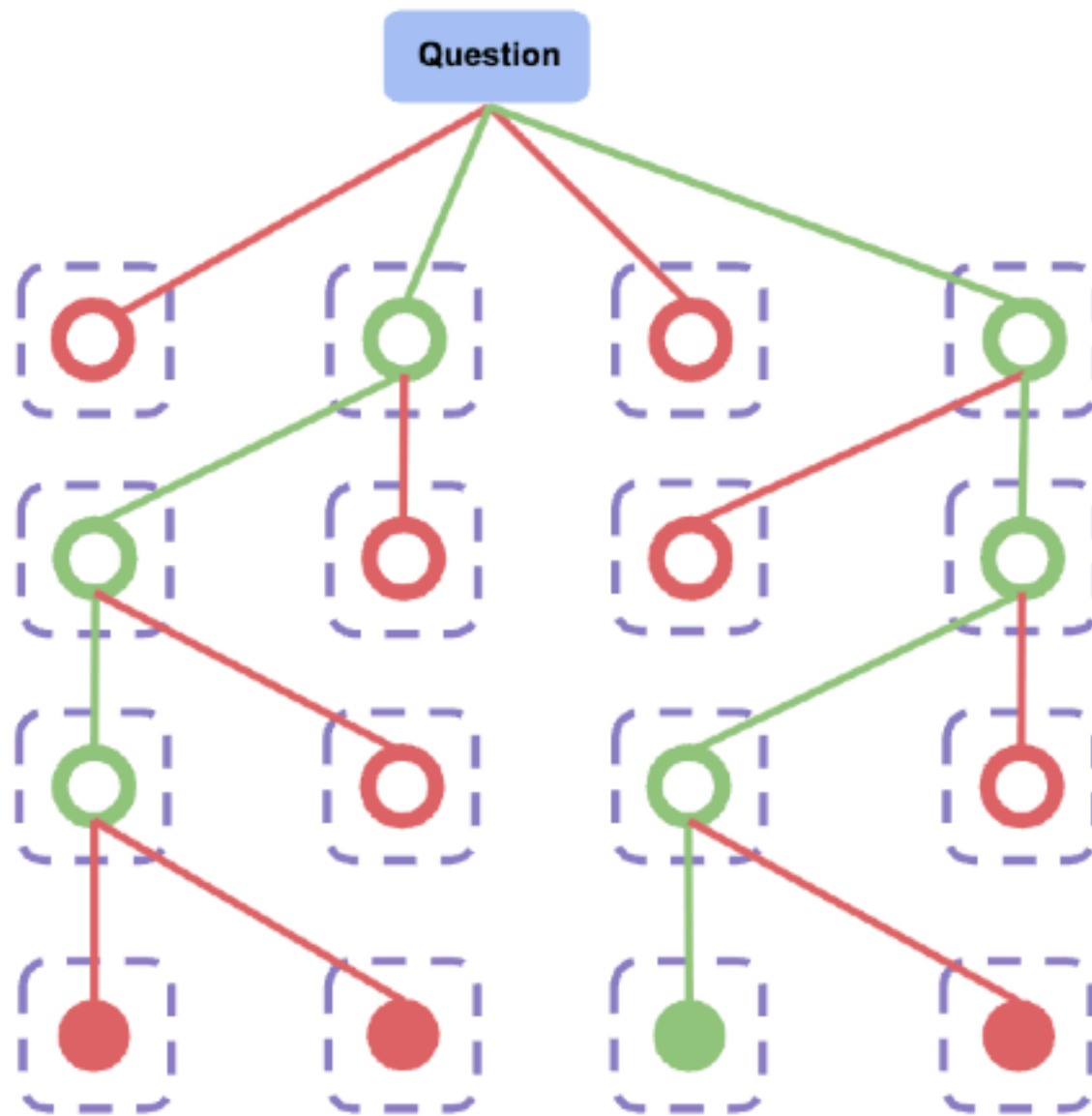
Best-of-N



Select the best final answer using the verifier

TAP

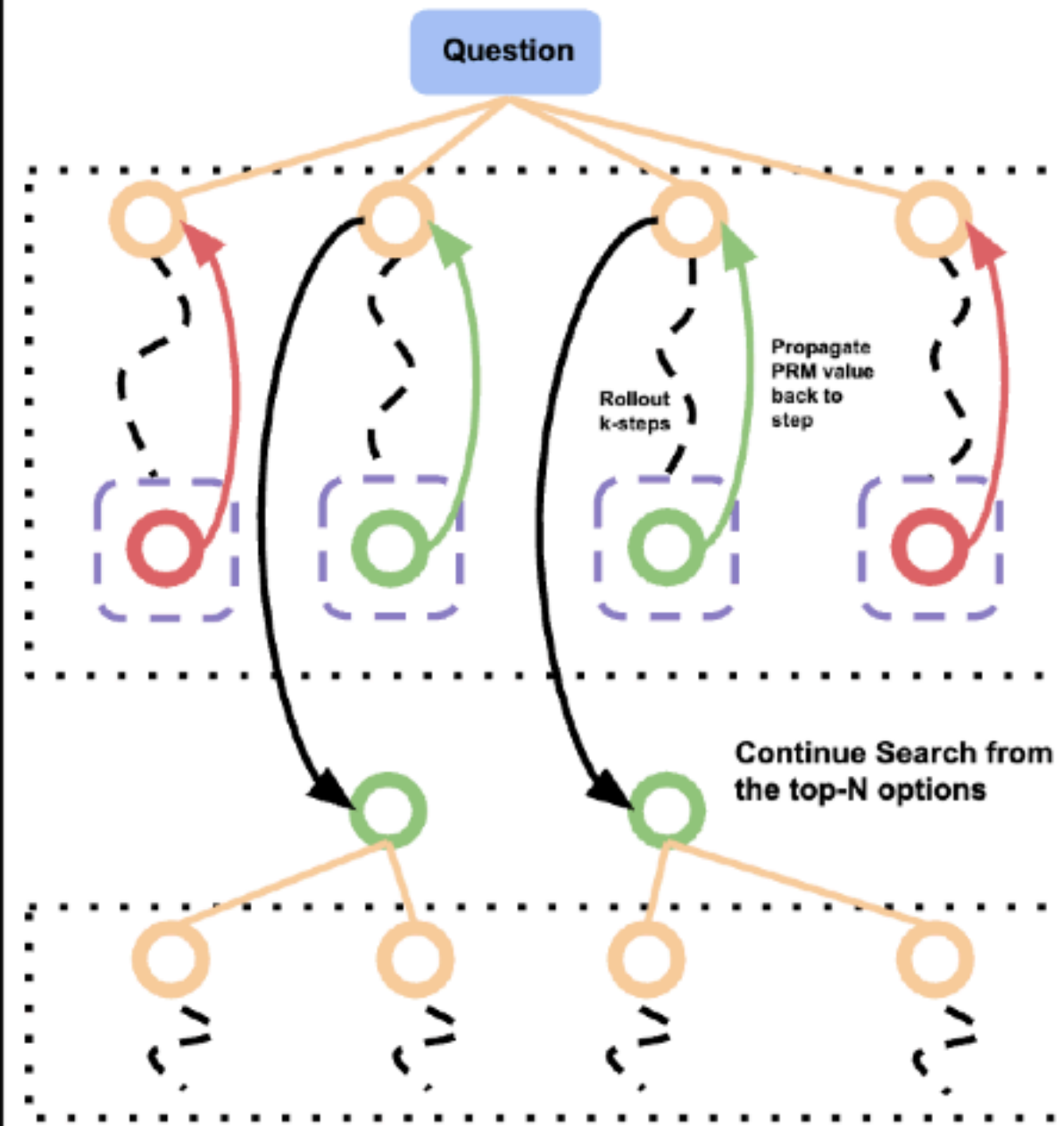
Tree search



Select the best final answer using the verifier

Adversarial reasoning

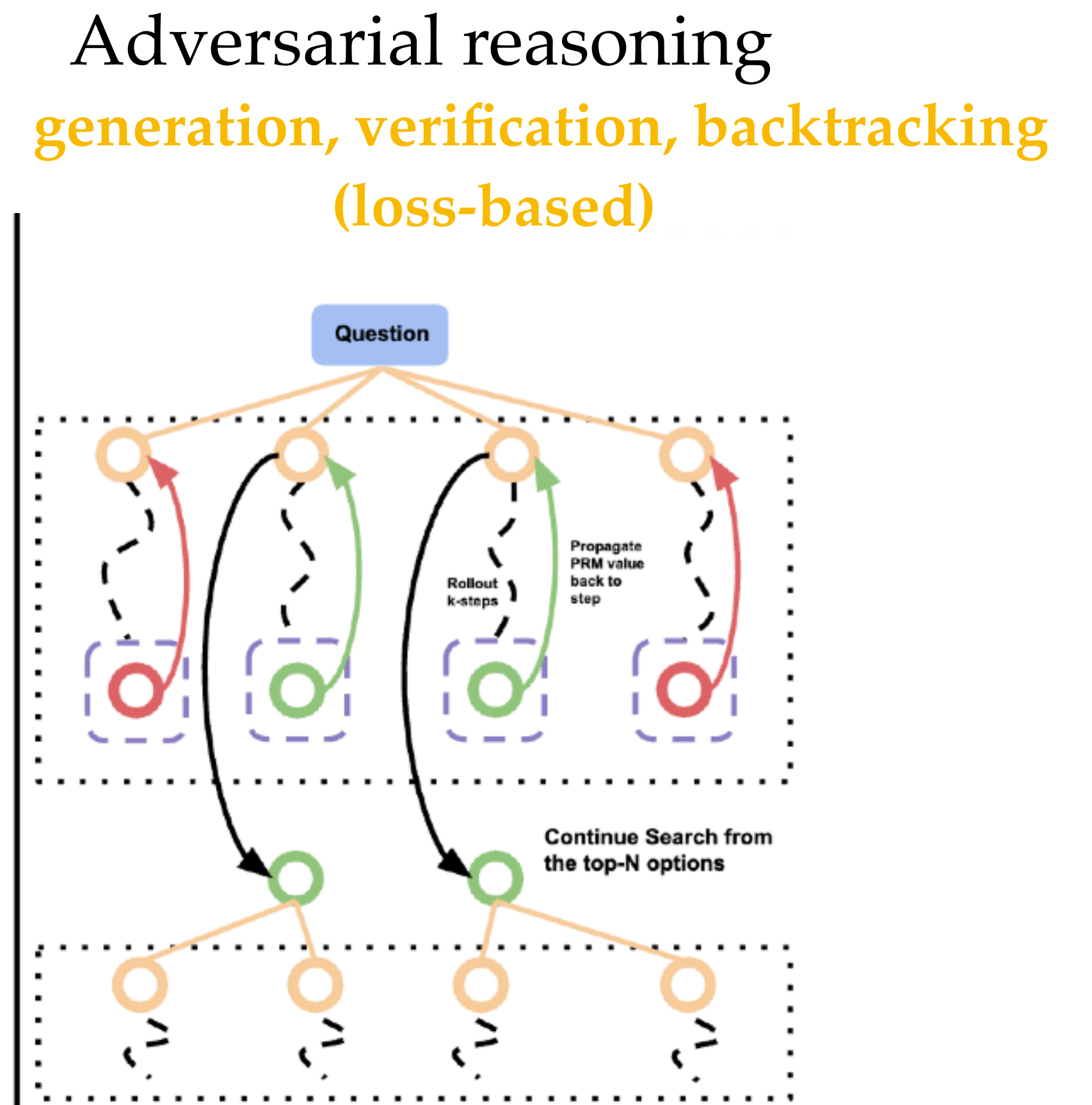
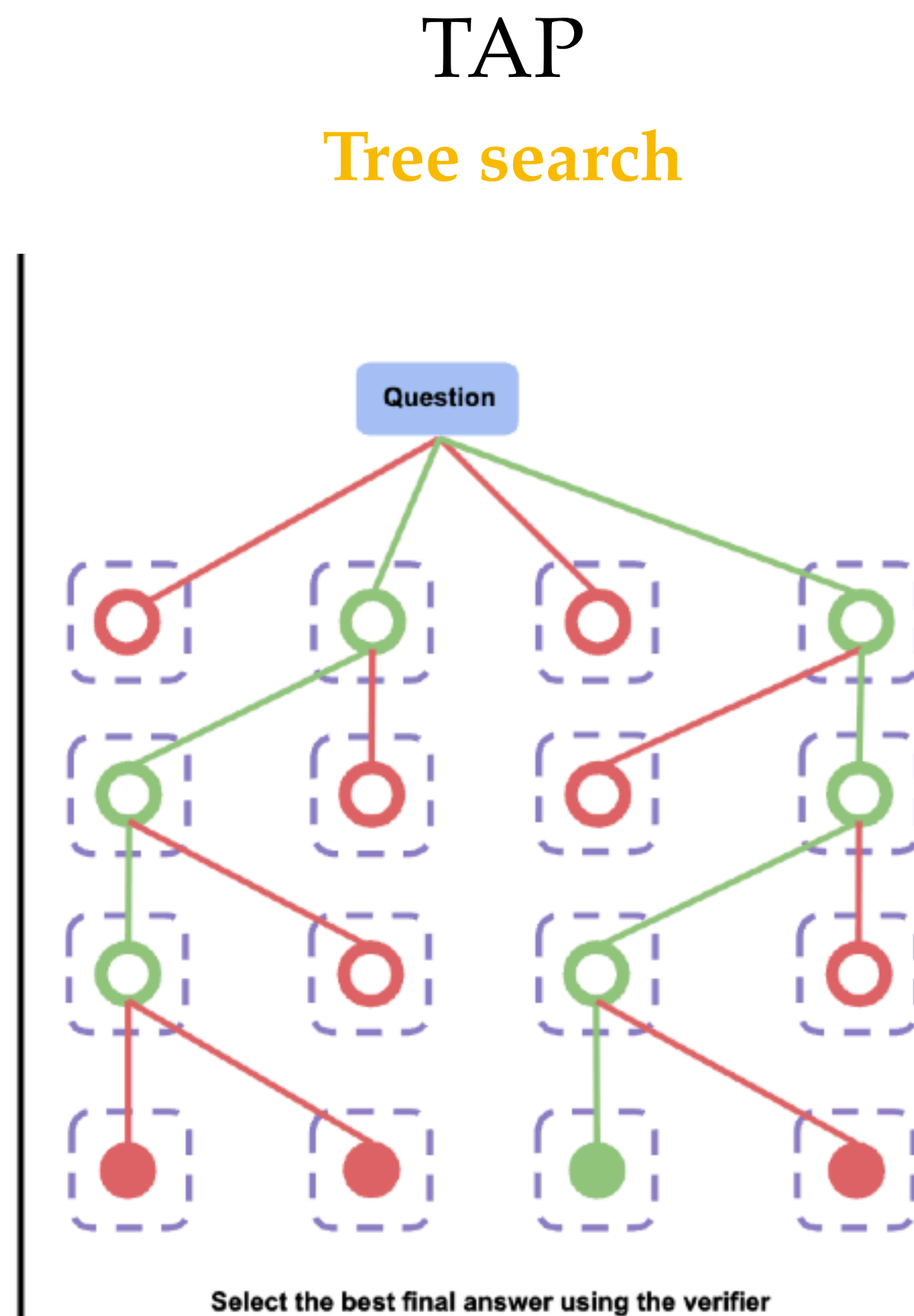
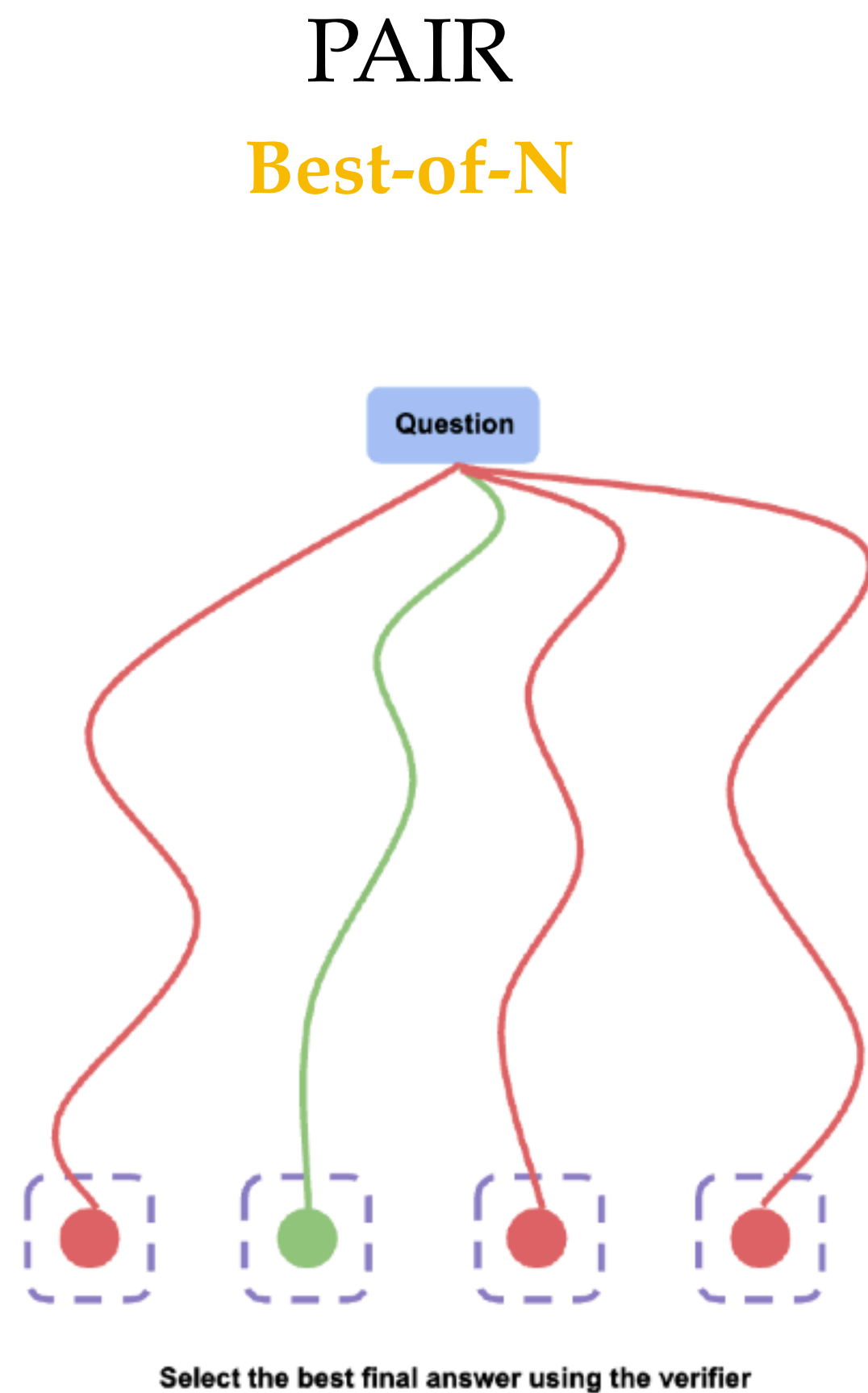
generation, verification, backtracking
(loss-based)

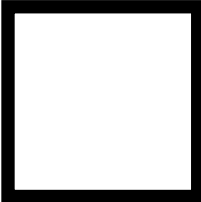
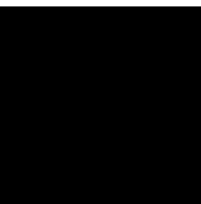


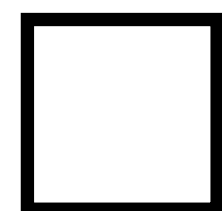
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Perspective: Jailbreaking as a search problem.



Algorithm	Search space	Threat model	Query Complexity
GCG	Token		High
Adaptive Attack	Token		High
PAIR	Prompt		Low
TAP	Prompt		Low
Adv Reasoning	Prompt		Low



: WhiteBox, needs gradient (also transfers to BlackBox models)



: GrayBox, needs loss values (also transfers to BlackBox models)



: BlackBox